

FIELD THEORY
B.E. Electrical Engineering
Second Year – Second Semester
2022 Examination – Part I

Complete Semester-Style Solution

Fast Compile Version: No physics package, no tikz-3dplot, no heavy shadows

How to Use This Document

This file solves the full Part-I question paper shown in the supplied image.

The answer style is designed for semester writing.

Every long answer follows the sequence:

definition → diagram → derivation → physical meaning → final boxed result

This code is intentionally kept free from the `physics` package.

This avoids command conflicts such as `\divergence` already defined.

This code also avoids `tdplot_main_coords`.

This avoids the TikZ error that appeared earlier.

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1 Question 1(a): Fundamental Definition of Curl and Cartesian Expression

1(a) What is the fundamental definition of curl of a vector field? Derive the expression for curl of a vector field $\mathbf{M}$, written as $\nabla \times \mathbf{M}$ where ∇ is the vector differential operator in Cartesian coordinates. What is the physical significance of curl?

1.1 Short Answer-Building Plan

1. First define curl using circulation per unit area.
2. Then take a small rectangular loop.
3. Derive one component of curl.
4. Then write other components by cyclic symmetry.
5. Finally write determinant form.
6. End with physical meaning.

1.2 Fundamental Definition

Curl of a vector field measures the circulation of the field per unit area.

For a vector field $\mathbf{M}$, the component of curl along unit normal $\hat{\mathbf{n}}$ is defined as

$$(\nabla \times \mathbf{M}) \cdot \hat{\mathbf{n}} = \lim_{\Delta S \rightarrow 0} \frac{1}{\Delta S} \oint_C \mathbf{M} \cdot d\mathbf{l}$$

Here C is the closed boundary of the small surface ΔS .

The symbol $\oint_C$ means line integration around a closed path.

The term $\mathbf{M} \cdot d\mathbf{l}$ means component of $\mathbf{M}$ along the path.

The quantity $\oint_C \mathbf{M} \cdot d\mathbf{l}$ is called circulation.

Therefore curl means circulation per unit area.

1.3 Derivation of Cartesian Curl

Let the vector field be

$$\mathbf{M} = M_x \hat{\mathbf{x}} + M_y \hat{\mathbf{y}} + M_z \hat{\mathbf{z}}.$$

We first derive the z -component of curl.

For this, take a small rectangle in the xy -plane.

The normal direction of this rectangle is $+\hat{\mathbf{z}}$.

The sides of the rectangle are dx and dy .

The area is

$$dS = dx dy.$$

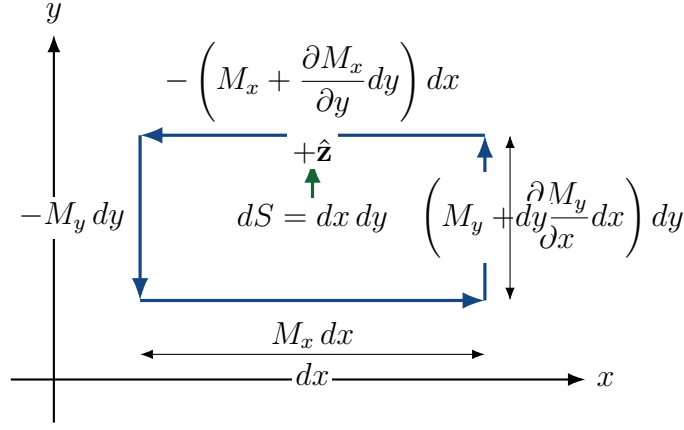


Figure 1: Small rectangular path in the xy -plane for deriving the z -component of curl.

Now calculate circulation around four sides.

Along bottom side:

$$\int_1 \mathbf{M} \cdot d\mathbf{l} = M_x dx.$$

Along right side:

$$\int_2 \mathbf{M} \cdot d\mathbf{l} = \left(M_y + \frac{\partial M_y}{\partial x} dx \right) dy.$$

Along top side, motion is in negative x direction:

$$\int_3 \mathbf{M} \cdot d\mathbf{l} = - \left(M_x + \frac{\partial M_x}{\partial y} dy \right) dx.$$

Along left side, motion is in negative y direction:

$$\int_4 \mathbf{M} \cdot d\mathbf{l} = -M_y dy.$$

Therefore total circulation is

$$\oint_C \mathbf{M} \cdot d\mathbf{l} = M_x dx + \left(M_y + \frac{\partial M_y}{\partial x} dx \right) dy - \left(M_x + \frac{\partial M_x}{\partial y} dy \right) dx - M_y dy.$$

On simplifying,

$$\oint_C \mathbf{M} \cdot d\mathbf{l} = \left(\frac{\partial M_y}{\partial x} - \frac{\partial M_x}{\partial y} \right) dx dy.$$

Divide by area $dx dy$.

$$(\nabla \times \mathbf{M})_z = \frac{\partial M_y}{\partial x} - \frac{\partial M_x}{\partial y}.$$

Similarly, by taking rectangles in other coordinate planes,

$$(\nabla \times \mathbf{M})_x = \frac{\partial M_z}{\partial y} - \frac{\partial M_y}{\partial z}.$$

$$(\nabla \times \mathbf{M})_y = \frac{\partial M_x}{\partial z} - \frac{\partial M_z}{\partial x}.$$

Therefore,

$$\nabla \times \mathbf{M} = \left(\frac{\partial M_z}{\partial y} - \frac{\partial M_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial M_x}{\partial z} - \frac{\partial M_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial M_y}{\partial x} - \frac{\partial M_x}{\partial y} \right) \hat{\mathbf{z}}$$

This is also written in determinant form as

$$\nabla \times \mathbf{M} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ M_x & M_y & M_z \end{vmatrix}$$

1.4 Physical Significance of Curl

Curl measures the local rotational tendency of a vector field.

If a small paddle wheel is placed in the field and it rotates, then curl exists at that point.

If the paddle wheel does not rotate, the curl is zero.

The direction of curl is given by the right-hand rule.

In electromagnetics, curl is very important.

For example,

$$\nabla \times \mathbf{H} = \mathbf{J}$$

means current density produces circulation of magnetic field intensity.

Similarly,

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

means time-varying magnetic field produces circulating electric field.

Curl measures circulation per unit area and physically represents local rotation of a vector field.

2 Question 1(b): Magnetic Boundary Relation

1(b) Establish the boundary relation for magnetic field when it passes through two different magnetic media having permeabilities μ_1 and μ_2 . Show one example for application of boundary relation for magnetic field.

2.1 Short Answer-Building Plan

1. Draw two media separated by an interface.
2. Use Gauss law for magnetism to get normal B condition.
3. Use Ampere law to get tangential H condition.
4. Combine with $B = \mu H$.
5. Obtain magnetic refraction relation.
6. Give transformer core or magnetic shielding as application.

2.2 Basic Situation

Consider two magnetic media separated by a plane boundary.

Medium 1 has permeability μ_1 .

Medium 2 has permeability μ_2 .

The magnetic field passes from medium 1 to medium 2.

The magnetic flux density and magnetic field intensity are related by

$$\mathbf{B}_1 = \mu_1 \mathbf{H}_1$$

and

$$\mathbf{B}_2 = \mu_2 \mathbf{H}_2.$$

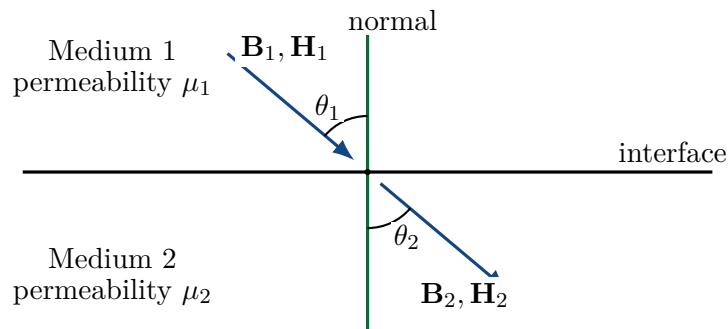


Figure 2: Magnetic field passing through two magnetic media.

2.3 Normal Component of Magnetic Flux Density

Gauss law for magnetism states that

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = 0.$$

This means net magnetic flux leaving any closed surface is zero.

Take a small pillbox across the boundary.

Let the cross-sectional area be ΔS .

As the height of the pillbox tends to zero, flux through side surface becomes zero.

So only top and bottom faces contribute.

Therefore,

$$B_{1n}\Delta S - B_{2n}\Delta S = 0.$$

Hence,

$$\boxed{B_{1n} = B_{2n}}$$

Therefore normal component of magnetic flux density is continuous.

Using $B = \mu H$,

$$\mu_1 H_{1n} = \mu_2 H_{2n}.$$

2.4 Tangential Component of Magnetic Field Intensity

Ampere's circuital law is

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}}.$$

Take a small rectangular loop across the boundary.

If there is no surface current at the boundary,

$$I_{\text{enc}} = 0.$$

As the height of the rectangle tends to zero, the contribution of the vertical sides becomes zero.

So,

$$H_{1t}\Delta l - H_{2t}\Delta l = 0.$$

Hence,

$$\boxed{H_{1t} = H_{2t}}$$

Therefore tangential component of magnetic field intensity is continuous when surface current is absent.

2.5 Magnetic Refraction Relation

Let θ_1 and θ_2 be angles made with the normal.

Normal condition:

$$B_1 \cos \theta_1 = B_2 \cos \theta_2.$$

Tangential condition:

$$H_1 \sin \theta_1 = H_2 \sin \theta_2.$$

Using $B = \mu H$,

$$H_1 = \frac{B_1}{\mu_1}$$

and

$$H_2 = \frac{B_2}{\mu_2}.$$

So tangential condition becomes

$$\frac{B_1}{\mu_1} \sin \theta_1 = \frac{B_2}{\mu_2} \sin \theta_2.$$

Divide this equation by the normal condition

$$B_1 \cos \theta_1 = B_2 \cos \theta_2.$$

Then

$$\frac{\tan \theta_1}{\mu_1} = \frac{\tan \theta_2}{\mu_2}.$$

Therefore,

$$\boxed{\frac{\tan \theta_1}{\tan \theta_2} = \frac{\mu_1}{\mu_2}}$$

2.6 Application

A direct application is in transformer iron cores.

Iron has very high permeability compared with air.

So magnetic flux prefers to remain inside iron.

This is why transformer cores guide magnetic flux and reduce leakage flux.

$$\boxed{B_{1n} = B_{2n}, \quad H_{1t} = H_{2t}, \quad \frac{\tan \theta_1}{\tan \theta_2} = \frac{\mu_1}{\mu_2}}$$

3 Question 2(a): Magnetic Vector Potential and Biot–Savart Law

2(a) What do you understand by magnetic vector potential? Deduce Biot–Savart law.

3.1 Magnetic Vector Potential

In magnetostatics, magnetic flux density satisfies

$$\nabla \cdot \mathbf{B} = 0.$$

This means magnetic field has no source or sink.

A vector field whose divergence is zero can be expressed as curl of another vector field.

Therefore we define magnetic vector potential $\mathbf{A}$ by

$$\boxed{\mathbf{B} = \nabla \times \mathbf{A}}$$

Here $\mathbf{A}$ is called magnetic vector potential.

The unit of $\mathbf{A}$ is Weber per metre.

It is useful because sometimes it is easier to calculate $\mathbf{A}$ first and then obtain $\mathbf{B}$ by taking curl.

3.2 Deduction of Biot–Savart Law

Ampere’s law in magnetostatics is

$$\nabla \times \mathbf{B} = \mu \mathbf{J}.$$

Using

$$\mathbf{B} = \nabla \times \mathbf{A},$$

we get

$$\nabla \times (\nabla \times \mathbf{A}) = \mu \mathbf{J}.$$

Use vector identity

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}.$$

For magnetostatics we use Coulomb gauge:

$$\nabla \cdot \mathbf{A} = 0.$$

Therefore,

$$-\nabla^2 \mathbf{A} = \mu \mathbf{J}.$$

So,

$$\boxed{\nabla^2 \mathbf{A} = -\mu \mathbf{J}}$$

This is Poisson equation for magnetic vector potential.

The solution is

$$\mathbf{A} = \frac{\mu}{4\pi} \int \frac{\mathbf{J} dv}{R}.$$

For a filamentary current,

$$\mathbf{J} dv = I d\mathbf{l}.$$

Therefore,

$$\mathbf{A} = \frac{\mu I}{4\pi} \oint \frac{d\mathbf{l}}{R}.$$

Now,

$$\mathbf{B} = \nabla \times \mathbf{A}.$$

So,

$$\mathbf{B} = \nabla \times \left[\frac{\mu I}{4\pi} \oint \frac{d\mathbf{l}}{R} \right].$$

Taking curl gives

$$d\mathbf{B} = \frac{\mu I}{4\pi} \frac{d\mathbf{l} \times \hat{\mathbf{R}}}{R^2}.$$

Therefore total magnetic flux density is

$$\boxed{\mathbf{B} = \frac{\mu I}{4\pi} \oint_C \frac{d\mathbf{l} \times \hat{\mathbf{R}}}{R^2}}$$

This is Biot–Savart law.

In terms of $\mathbf{R}$,

$$\boxed{d\mathbf{B} = \frac{\mu I}{4\pi} \frac{d\mathbf{l} \times \mathbf{R}}{R^3}}$$

Since $\mathbf{B} = \mu \mathbf{H}$,

$$\boxed{d\mathbf{H} = \frac{I}{4\pi} \frac{d\mathbf{l} \times \hat{\mathbf{R}}}{R^2}}$$

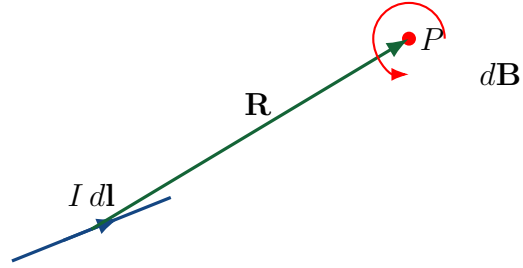


Figure 3: Current element producing magnetic field at point P .

$$\mathbf{B} = \frac{\mu I}{4\pi} \oint_C \frac{d\mathbf{l} \times \hat{\mathbf{R}}}{R^2}$$

4 Question 2(b): Rectangular Loop Numerical

2(b) A thin ring of rectangular loop having sides of 4 cm by 8 cm carries 50 A. Find H at a point on the axis positioned at a height of 10 cm on the axis of the loop. Derive the formula used.

4.1 Given Data

Side lengths are

$$4 \text{ cm} \times 8 \text{ cm}.$$

So half sides are

$$a = 2 \text{ cm} = 0.02 \text{ m}$$

and

$$b = 4 \text{ cm} = 0.04 \text{ m}.$$

Current is

$$I = 50 \text{ A}.$$

Axial distance is

$$h = 10 \text{ cm} = 0.10 \text{ m}.$$

4.2 Formula for Rectangular Loop

For a rectangular loop with half sides a and b , carrying current I , magnetic field intensity on the axis at height h is

$$H = \frac{Iab}{\pi\sqrt{a^2 + b^2 + h^2}} \left[\frac{1}{a^2 + h^2} + \frac{1}{b^2 + h^2} \right]$$

The direction is along the axis of the loop.

4.3 Derivation Outline

Biot-Savart law for magnetic field intensity is

$$d\mathbf{H} = \frac{I}{4\pi} \frac{d\mathbf{l} \times \hat{\mathbf{R}}}{R^2}.$$

For a finite straight conductor, the magnetic field intensity at a point is

$$H = \frac{I}{4\pi\rho} (\sin \alpha_1 + \sin \alpha_2).$$

For the rectangle, contributions from four sides are added.

Due to symmetry, horizontal components cancel.

Only axial components add.

For the pair of sides parallel to y -axis,

$$H_{z1} = \frac{Iab}{\pi(a^2 + h^2)\sqrt{a^2 + b^2 + h^2}}.$$

For the pair of sides parallel to x -axis,

$$H_{z2} = \frac{Iab}{\pi(b^2 + h^2)\sqrt{a^2 + b^2 + h^2}}.$$

Therefore total axial field is

$$H = H_{z1} + H_{z2}.$$

Hence,

$$H = \frac{Iab}{\pi\sqrt{a^2 + b^2 + h^2}} \left[\frac{1}{a^2 + h^2} + \frac{1}{b^2 + h^2} \right].$$

4.4 Numerical Calculation

Calculate squared quantities:

$$a^2 = (0.02)^2 = 0.0004.$$

$$b^2 = (0.04)^2 = 0.0016.$$

$$h^2 = (0.10)^2 = 0.0100.$$

Now,

$$a^2 + h^2 = 0.0004 + 0.0100 = 0.0104.$$

$$b^2 + h^2 = 0.0016 + 0.0100 = 0.0116.$$

$$a^2 + b^2 + h^2 = 0.0004 + 0.0016 + 0.0100 = 0.0120.$$

Therefore,

$$\sqrt{a^2 + b^2 + h^2} = \sqrt{0.0120} = 0.10954.$$

Now,

$$Iab = 50(0.02)(0.04) = 0.04.$$

The bracket term is

$$\left[\frac{1}{0.0104} + \frac{1}{0.0116} \right] = 96.1538 + 86.2069.$$

$$= 182.3607.$$

Thus,

$$H = \frac{0.04}{\pi(0.10954)}(182.3607).$$

$$H = 0.11623(182.3607).$$

$$H = 21.20 \text{ A/m}.$$

$$\boxed{H \approx 21.2 \text{ A/m}}$$

The direction is along the axis of the loop, depending on the direction of current by right-hand rule.

5 Question 3(a): Ampere–Maxwell Law and Displacement Current

3(a) Establish $\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$ and show the pictorial view of this relation. What is displacement current?

5.1 Old Ampere Law

For steady current, Ampere law is

$$\nabla \times \mathbf{H} = \mathbf{J}.$$

This law says current density produces circulation of magnetic field intensity.

But this law is valid only for steady current.

It fails in time-varying fields.

5.2 Need for Maxwell Correction

Take divergence of both sides of old Ampere law:

$$\nabla \cdot (\nabla \times \mathbf{H}) = \nabla \cdot \mathbf{J}.$$

But divergence of curl is always zero:

$$\nabla \cdot (\nabla \times \mathbf{H}) = 0.$$

Therefore old law gives

$$\nabla \cdot \mathbf{J} = 0.$$

But continuity equation is

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}.$$

For time-varying charge density,

$$\frac{\partial \rho_v}{\partial t} \neq 0.$$

So old Ampere law contradicts charge conservation.

Maxwell corrected this by adding displacement current density.

5.3 Displacement Current Density

Gauss law is

$$\nabla \cdot \mathbf{D} = \rho_v.$$

Differentiate with respect to time:

$$\nabla \cdot \frac{\partial \mathbf{D}}{\partial t} = \frac{\partial \rho_v}{\partial t}.$$

From continuity equation,

$$\frac{\partial \rho_v}{\partial t} = -\nabla \cdot \mathbf{J}.$$

So,

$$\nabla \cdot \frac{\partial \mathbf{D}}{\partial t} = -\nabla \cdot \mathbf{J}.$$

Therefore,

$$\nabla \cdot \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \right) = 0.$$

So Maxwell modified Ampere law as

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

Here,

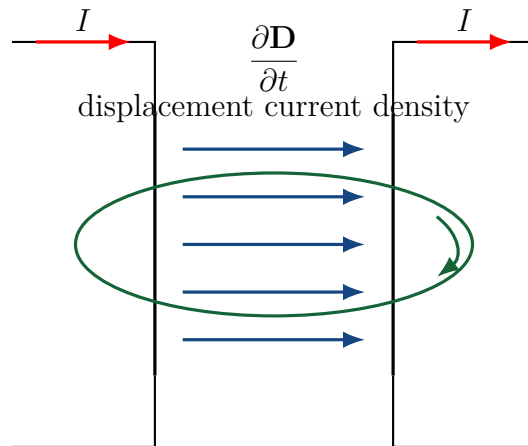
$$\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t}$$

is called displacement current density.

The total current density is

$$\mathbf{J}_{total} = \mathbf{J} + \mathbf{J}_d.$$

5.4 Pictorial View



Changing electric field between plates acts like current

Figure 4: Displacement current between capacitor plates.

5.5 Physical Meaning

Displacement current is not conduction current due to actual flow of free charges.

It is a current-like effect produced by a time-varying electric flux density.

It allows magnetic field to exist between capacitor plates even though no conduction current crosses the dielectric.

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

$$\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t}$$

6 Question 3(b): Faraday Law

3(b) Derive $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$ and show the pictorial view of the relation.

6.1 Faraday Law in Integral Form

Faraday law states that induced emf around a closed path is equal to the negative rate of change of magnetic flux linked with that path.

Mathematically,

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{S}.$$

For a fixed surface,

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}.$$

6.2 Using Stokes' Theorem

Stokes' theorem states that

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S}.$$

Therefore,

$$\int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}.$$

Since the surface is arbitrary,

$$\boxed{\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}}$$

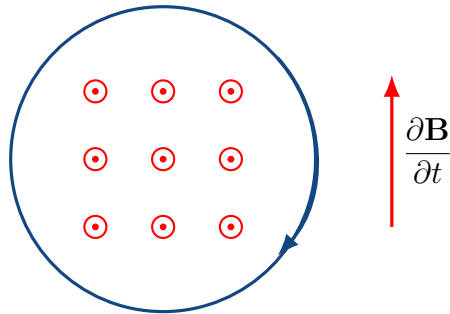
6.3 Meaning of Negative Sign

The negative sign represents Lenz's law.

It means induced electric field opposes the cause producing it.

6.4 Pictorial View

Changing magnetic field



Induced circulating electric field $\mathbf{E}$

Figure 5: Changing magnetic field produces circulating electric field.

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

A time-varying magnetic field produces circulating electric field.

7 Question 4(a): Electromagnetic Wave Equations and Plane Wave Solution

4(a) Derive electromagnetic wave equations from Maxwell's equations. Using E.M. wave equation in free space, obtain analytical solution of the wave equation of Electric Field $\mathbf{E}$ considering it as a plane wave and also draw the wave propagation.

7.1 Maxwell Curl Equations

For a homogeneous conducting medium,

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}.$$

Since

$$\mathbf{B} = \mu \mathbf{H},$$

we get

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t}.$$

Also,

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}.$$

Using

$$\mathbf{J} = \sigma \mathbf{E}$$

and

$$\mathbf{D} = \epsilon \mathbf{E},$$

we get

$$\nabla \times \mathbf{H} = \sigma \mathbf{E} + \epsilon \frac{\partial \mathbf{E}}{\partial t}.$$

7.2 Wave Equation for Electric Field

Take curl of Faraday equation:

$$\nabla \times (\nabla \times \mathbf{E}) = -\mu \frac{\partial}{\partial t} (\nabla \times \mathbf{H}).$$

Substitute

$$\nabla \times \mathbf{H} = \sigma \mathbf{E} + \epsilon \frac{\partial \mathbf{E}}{\partial t}.$$

Then

$$\nabla \times (\nabla \times \mathbf{E}) = -\mu\sigma \frac{\partial \mathbf{E}}{\partial t} - \mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}.$$

Use vector identity

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}.$$

In source-free region,

$$\nabla \cdot \mathbf{E} = 0.$$

Therefore,

$$-\nabla^2 \mathbf{E} = -\mu\sigma \frac{\partial \mathbf{E}}{\partial t} - \mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}.$$

Hence,

$$\boxed{\nabla^2 \mathbf{E} = \mu\sigma \frac{\partial \mathbf{E}}{\partial t} + \mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}}$$

This is the electric field wave equation in conducting medium.

7.3 Wave Equation for Magnetic Field

Similarly,

$$\boxed{\nabla^2 \mathbf{H} = \mu\sigma \frac{\partial \mathbf{H}}{\partial t} + \mu\epsilon \frac{\partial^2 \mathbf{H}}{\partial t^2}}$$

7.4 Free-Space Wave Equation

For free space,

$$\sigma = 0, \quad \mu = \mu_0, \quad \epsilon = \epsilon_0.$$

So,

$$\boxed{\nabla^2 \mathbf{E} = \mu_0\epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}}$$

and

$$\boxed{\nabla^2 \mathbf{H} = \mu_0\epsilon_0 \frac{\partial^2 \mathbf{H}}{\partial t^2}}$$

7.5 Plane Wave Solution for Electric Field

Let the wave travel along $+x$ direction.

Let electric field be along y direction.

So,

$$\mathbf{E} = E_y(x, t)\hat{\mathbf{y}}.$$

For a plane wave,

$$\frac{\partial}{\partial y} = 0$$

and

$$\frac{\partial}{\partial z} = 0.$$

Therefore the wave equation becomes

$$\frac{\partial^2 E_y}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 E_y}{\partial t^2}.$$

The standard sinusoidal solution is

$$E_y(x, t) = E_0 \cos(\omega t - \beta x)$$

where

$$\beta = \omega \sqrt{\mu_0 \epsilon_0}.$$

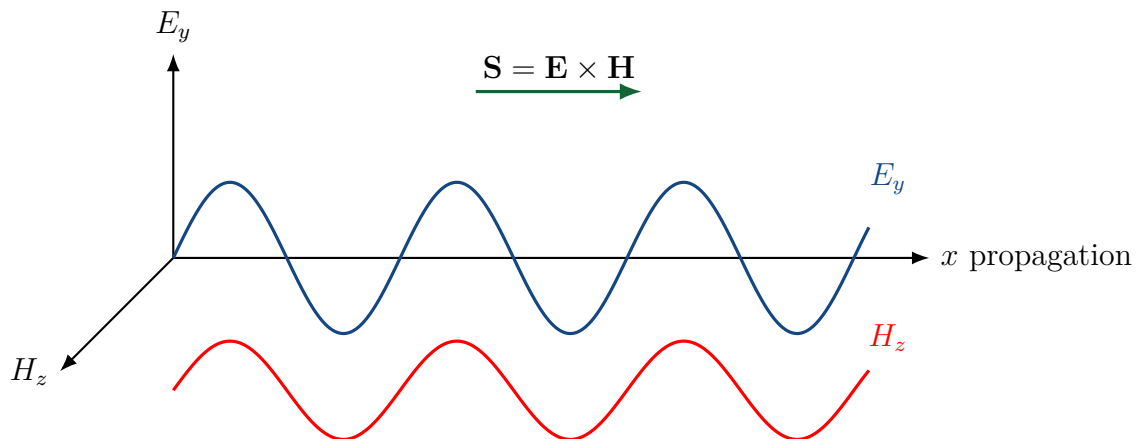
Wave velocity is

$$v = \frac{\omega}{\beta} = \frac{1}{\sqrt{\mu_0 \epsilon_0}}.$$

In free space,

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \approx 3 \times 10^8 \text{ m/s}$$

7.6 Wave Propagation Diagram



Plane EM wave: $\mathbf{E}$, $\mathbf{H}$ and propagation direction are mutually perpendicular

Figure 6: Plane electromagnetic wave propagation in free space.

$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

$$E_y(x, t) = E_0 \cos(\omega t - \beta x)$$

8 Question 4(b): Electromagnetic Wave Polarization

4(b) Describe electromagnetic wave polarizations.

8.1 Definition

Polarization describes the time variation of the direction of electric field vector of an electromagnetic wave at a fixed point in space.

Although an EM wave contains both electric field and magnetic field, polarization is described by the electric field vector $\mathbf{E}$.

The path traced by the tip of $\mathbf{E}$ decides the type of polarization.

8.2 Linear Polarization

If the electric field vector remains along a fixed straight line, the wave is linearly polarized.

For example,

$$\mathbf{E} = E_0 \cos(\omega t - \beta z) \hat{\mathbf{x}}.$$

Here the electric field is always along x direction.

Only its magnitude changes with time.

So the wave is linearly polarized.

8.3 Circular Polarization

If the electric field has two perpendicular components of equal amplitude and phase difference of 90° , then the tip of electric field vector traces a circle.

Conditions are

$$E_{0x} = E_{0y}$$

and

$$\delta = \pm \frac{\pi}{2}.$$

Then the wave is circularly polarized.

8.4 Elliptical Polarization

If the two perpendicular components have unequal amplitudes or a general phase difference, then the tip of electric field vector traces an ellipse.

This is called elliptical polarization.

Linear and circular polarization are special cases of elliptical polarization.

Path traced by the tip of electric field vector

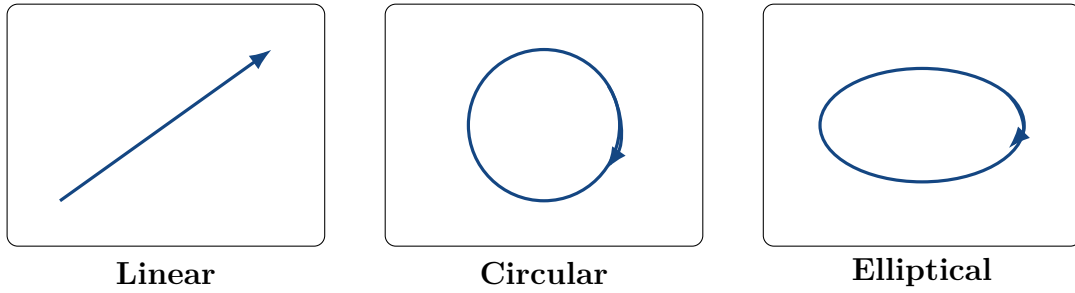


Figure 7: Three basic types of EM wave polarization.

Polarization types: linear, circular and elliptical.

9 Question 5: Short Notes

5. Write short notes on any four: (a) $\nabla \cdot \mathbf{J} = 0$ (b) $\nabla \cdot \mathbf{B} = 0$ (c) EM plane wave propagation (d) $\nabla \times \mathbf{H} = \mathbf{J}$ (e) Stokes' theorem (f) Poynting vector.

9.1 Short Note (a): $\nabla \cdot \mathbf{J} = 0$

The equation

$$\nabla \cdot \mathbf{J} = 0$$

represents steady current condition.

The general continuity equation is

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}.$$

For steady current,

$$\frac{\partial \rho_v}{\partial t} = 0.$$

Therefore,

$$\nabla \cdot \mathbf{J} = 0.$$

Physical meaning:

No charge accumulation takes place at any point.

Current entering a small volume equals current leaving the volume.

This is the field form of Kirchhoff's current law.

9.2 Short Note (b): $\nabla \cdot \mathbf{B} = 0$

The equation

$$\nabla \cdot \mathbf{B} = 0$$

is Gauss law for magnetism in differential form.

It means magnetic flux density has no source or sink.

Magnetic field lines are closed loops.

There are no isolated magnetic monopoles.

Integral form is

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = 0$$

This means net magnetic flux through any closed surface is zero.

9.3 Short Note (c): EM Plane Wave Propagation

An electromagnetic plane wave consists of time-varying electric and magnetic fields.

In free space,

$$\mathbf{E} \perp \mathbf{H}$$

and both are perpendicular to direction of propagation.

For wave travelling along $+x$ direction,

$$\mathbf{E} = E_y \hat{\mathbf{y}}$$

and

$$\mathbf{H} = H_z \hat{\mathbf{z}}.$$

The direction of propagation is

$$\mathbf{E} \times \mathbf{H}.$$

The velocity in free space is

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

9.4 Short Note (d): $\nabla \times \mathbf{H} = \mathbf{J}$

The equation

$$\nabla \times \mathbf{H} = \mathbf{J}$$

is Ampere's law in differential form for magnetostatics.

It means current density is the source of circulation of magnetic field intensity.

Integral form is

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}}.$$

Using

$$I_{\text{enc}} = \int_S \mathbf{J} \cdot d\mathbf{S}$$

and Stokes' theorem,

$$\int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S} = \int_S \mathbf{J} \cdot d\mathbf{S}.$$

So,

$$\nabla \times \mathbf{H} = \mathbf{J}.$$

9.5 Short Note (e): Stokes' Theorem

Stokes' theorem relates line integral around a closed boundary to surface integral of curl over the surface.

Mathematically,

$$\oint_C \mathbf{A} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

Left side represents circulation around closed path C .

Right side represents total curl passing through surface S .

It is used to convert integral form of Ampere law and Faraday law into differential form.

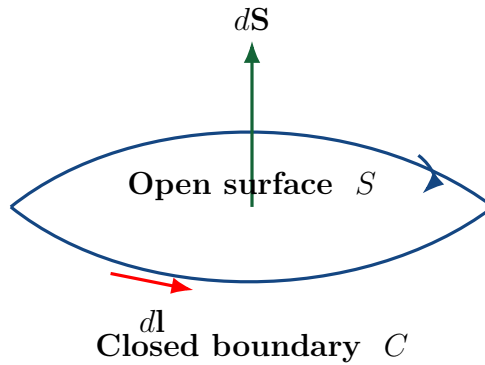


Figure 8: Stokes' theorem.

9.6 Short Note (f): Poynting Vector

Poynting vector represents electromagnetic power flow per unit area.

It is defined as

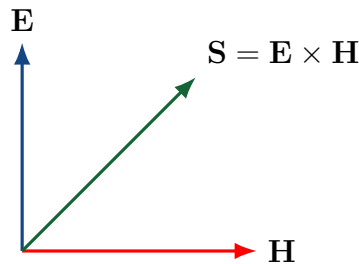
$$\mathbf{S} = \mathbf{E} \times \mathbf{H}$$

Its unit is

$$W/m^2.$$

The direction of $\mathbf{S}$ gives the direction of energy propagation.

For an electromagnetic wave, $\mathbf{E}$, $\mathbf{H}$ and $\mathbf{S}$ are mutually perpendicular.



Direction of EM energy flow

Figure 9: Poynting vector direction.

For short notes, always write definition, formula, unit if needed, and physical meaning.

10 Final Exam Strategy for This Paper

1. For Question 1(a), draw the rectangular curl diagram and write determinant form.
2. For Question 1(b), draw interface diagram and remember B_n continuous, H_t continuous.
3. For Question 2(a), begin from $\mathbf{B} = \nabla \times \mathbf{A}$ and end with Biot–Savart law.
4. For Question 2(b), convert centimetres to metres before calculation.
5. For Question 3(a), explain why displacement current is needed.
6. For Question 3(b), use Faraday integral law and Stokes’ theorem.
7. For Question 4(a), start from Maxwell curl equations and derive wave equation.
8. For Question 4(b), explain polarization by path of tip of electric field vector.
9. For Question 5, write short notes in definition–formula–meaning format.

11 Final Formula Sheet

$$\nabla \times \mathbf{M} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ M_x & M_y & M_z \end{vmatrix}$$

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\mu_1}{\mu_2}$$

$$\mathbf{B} = \nabla \times \mathbf{A}$$

$$d\mathbf{B} = \frac{\mu I}{4\pi} \frac{d\mathbf{l} \times \hat{\mathbf{R}}}{R^2}$$

$$H = \frac{Iab}{\pi\sqrt{a^2 + b^2 + h^2}} \left[\frac{1}{a^2 + h^2} + \frac{1}{b^2 + h^2} \right]$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla^2 \mathbf{E} = \mu\sigma \frac{\partial \mathbf{E}}{\partial t} + \mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

$$\nabla^2 \mathbf{E} = \mu_0\epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

$$E_y(x, t) = E_0 \cos(\omega t - \beta x)$$

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}$$

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$